Deepak Arun P

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About

Final-year Mechatronics Engineering student with a strong foundation in mechanical design, electrical systems and control engineering. Experienced in CAD modelling, structural analysis, motor control, MATLAB/Simulink and industrial automation, with practical exposure to automotive manufacturing and multidisciplinary system integration.

Education

Vellore Institute of Technology, B.Tech Mechatronics & Automation 2023 – 2027
CGPA: 8.49

Vidyaa Vikas, Karamadai, Higher Secondary (HSC) 2022
Grade: 80%

Vidyaa Vikas, Karamadai, Secondary (SSC) 2020
Grade: 88%

Technical Skills

Mechanical Design:	SolidWorks, Fusion 360, AutoCAD, CAD Modelling
Electrical & Electronics:	DC Motors, Sensors, Actuators, Motor Drivers, LTSpice
Control & Automation:	MATLAB/Simulink, PID Control, LabVIEW, PLC Programming, Ladder Logic, HMI
Simulation & Analysis:	ANSYS, FEA, Structural Analysis, LMS Amesim
Engineering Methods:	FMEA
Programming:	Python, Java

Internships

Hyundai Motor India Limited

- Performed FMEA of automated transmission assembly processes, identifying failure modes, evaluating Risk Priority Numbers (RPNs) and proposing corrective actions that reduced process risks by over 50%.
- Proposed PLC-based and mechanical poka-yoke improvements involving sensor integration, error detection logic and fixture modifications to prevent assembly defects.
- Gained practical exposure to transmission assembly, PLC-controlled machinery, ladder logic, sensors, actuators, machine interlocks and industrial automation safety.

Spark Minda, Sriperumbudur

- Worked in New Product Development (NPD), gaining exposure to automotive wiring harness design, prototype development and manufacturing processes.
- Gained exposure to product validation, electrical quality checks, industry standards and automotive product development workflows.

Projects

NASA Human Exploration Rover Challenge (HERC)

- Led rover subsystem CAD design, assembly modelling and structural validation using FEA, contributing to mechanical design optimisation and system integration.
- Designed and optimised rover wheels for road and off-road operation, while contributing to electrical power management and control systems.
- Coordinated mechanical, electrical and software teams for system integration, contributing to the rover achieving the fastest on-track performance during competition testing.

Compact Torque-Limiting Ball Detent Coupler

- Designed a compact ball-detent torque-limiting coupler for mechanical overload protection and controlled disengagement in small-scale machinery.
- Performed FEA to evaluate stress distribution, deformation and torque capacity, refining the design for reliable mechanical performance.

IIoT-Enabled Industrial Hopper Monitoring System

- Developed an IIoT-based hopper monitoring system using ESP32-CAM and ToF sensing for real-time material identification and fill-level measurement.
- Integrated image classification and Arduino Cloud for material monitoring, visualisation and low-material-level alerts.

Design, Assembly and PID Control of a Mini Robotic Arm

- Designed and assembled a 2-DOF planar robotic arm, and developed an Arduino-based LabVIEW HMI for DC motor speed and direction control with real-time IR sensor feedback.
- Modelled the robotic arm using MATLAB/Simulink and Simscape Multibody, implementing inverse kinematics and independent PID joint controllers to analyse positioning accuracy, response time, overshoot and steady-state error under gravity.

Comparative Analysis of Arc and MIG Welded Joints

- Evaluated and compared the mechanical properties of Arc and MIG welded joints through hardness, tensile strength and bend tests to assess weld strength, ductility and structural integrity.

Caterpillar Rover Competition

- Contributed to mechanical design, structural analysis, fabrication and subsystem integration for an off-road rover.

Languages

Tamil, English, Hindi